TWO MAPS BETWEEN ADMISSIBLE SEQUENCES WITH VANISHING OUTER COMPONENTS COMPOSE TO ZERO

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-triseq (source lines 4049-4076); the proof environment begins two lines below it and is its own proof, closing at source line 4098.

Editorial note. The statement refers to the author's standing Situation situation-ABC by label; that situation's text is retained as a context span and the generator expands the context transitively, so the reference resolves inside the assembled package.

Editorial note. The statement contains two displayed Xy-pic diagrams of squares with vanishing outer components; they are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: two maps between admissible sequences whose outer components vanish compose to a homotopically trivial map.

Allowed standard prerequisites (editorial).

- The standing situation (axioms A, B, C). A dg category with the distinguished-triangle axioms, including the admissible short exact sequence attached to a morphism and its structure maps (the author's Situation situation-ABC, retained as context).. The middle admissible sequence and its structure maps are used throughout the chase (source lines 4051-4053, 4078-4090).
- Commuting up to homotopy and vanishing outer components. The two squares commute up to homotopy and the maps on the outer terms are zero, so the given maps can be corrected by the homotopies without changing the situation.. The corrections produce the explicit families whose composites are then shown to vanish (source lines 4054-4076, 4078-4095).
- Admissible sequence structure maps. The inclusion, projection and splitting of an admissible short exact sequence and the identities among them.. The chase rewrites the maps using those structure maps and concludes from g b~ = 0 and b~' f = 0 (source lines 4085-4097).

Situation 0.1. Here R is a ring and $\mathcal{A}$ is a differential graded category over R having axioms (A), (B), and

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(C) given an arrow $f : x \to y$ of degree 0 with $\mathrm{d}(f) = 0$ there exists an admissible short exact sequence $y \to c(f) \to x[1]$ in $\mathrm{Comp}(\mathcal{A})$ such that the map $x[1] \to y[1]$ of Lemma 0.3 is equal to $f[1]$.

Lemma 0.2. *In Situation 0.1 given a diagram*

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ z & \xrightarrow{g} & w \end{array}$$

in Comp($\mathcal{A}$) commuting up to homotopy. Then

(1) *If f is an admissible monomorphism, then b is homotopic to a morphism b' which makes the diagram commute.*
(2) *If g is an admissible epimorphism, then a is homotopic to a morphism a' which makes the diagram commute.*

Proof. To prove (1), observe that the hypothesis implies that there is some $h \in \mathrm{Hom}_{\mathcal{A}}(x, w)$ of degree -1 such that $bf - ga = d(h)$. Since f is an admissible monomorphism, there is a morphism $\pi : y \to x$ in the category $\mathcal{A}$ of degree 0. Let $b' = b - d(h\pi)$. Then

$$\begin{aligned} b'f = bf - d(h\pi)f &= bf - d(h\pi f) \quad (\text{since } d(f) = 0) \\ &= bf - d(h) \\ &= ga \end{aligned}$$

as desired. The proof for (2) is omitted. □

Lemma 0.3. *Let $\mathcal{A}$ be a differential graded category satisfying axioms (A) and (B). Given an admissible short exact sequence $x \to y \to z$ we obtain (see proof) a triangle*

$$x \to y \to z \to x[1]$$

in Comp($\mathcal{A}$) with the property that any two compositions in $z[-1] \to x \to y \to z \to x[1]$ are zero in $K(\mathcal{A})$.

Proof. Choose a diagram

$$\begin{array}{ccccc} x & & \xrightarrow{\quad 1 \quad} & & x \\ & {\scriptstyle a}\searrow & & \nearrow{\scriptstyle \pi} & \\ & & y & & \\ & {\scriptstyle s}\nearrow & & \searrow{\scriptstyle b} & \\ z & & \xrightarrow{\quad 1 \quad} & & z \end{array}$$

giving the isomorphism of graded objects $y \cong x \oplus z$ as in the definition of an admissible short exact sequence. Here are some equations that hold in this situation

(1) $1 = \pi a$ and hence $\mathrm{d}(\pi)a = 0$,
(2) $1 = bs$ and hence $b\mathrm{d}(s) = 0$,
(3) $1 = a\pi + sb$ and hence $a\mathrm{d}(\pi) + \mathrm{d}(s)b = 0$,
(4) $\pi s = 0$ and hence $\mathrm{d}(\pi)s + \pi\mathrm{d}(s) = 0$,
(5) $\mathrm{d}(s) = a\pi\mathrm{d}(s)$ because $\mathrm{d}(s) = (a\pi + sb)\mathrm{d}(s)$ and $b\mathrm{d}(s) = 0$,
(6) $\mathrm{d}(\pi) = \mathrm{d}(\pi)sb$ because $\mathrm{d}(\pi) = \mathrm{d}(\pi)(a\pi + sb)$ and $\mathrm{d}(\pi)a = 0$,

(7) $\mathrm{d}(\pi\mathrm{d}(s)) = 0$ because if we postcompose it with the monomorphism a we get $\mathrm{d}(a\pi\mathrm{d}(s)) = \mathrm{d}(\mathrm{d}(s)) = 0$, and
(8) $\mathrm{d}(\mathrm{d}(\pi)s) = 0$ as by (4) it is the negative of $\mathrm{d}(\pi\mathrm{d}(s))$ which is 0 by (7).

We've used repeatedly that $\mathrm{d}(a) = 0$, $\mathrm{d}(b) = 0$, and that $\mathrm{d}(1) = 0$. By (7) we see that

$$\delta = \pi\mathrm{d}(s) = -\mathrm{d}(\pi)s : z \to x[1]$$

is a morphism in $\mathrm{Comp}(\mathcal{A})$. By (5) we see that the composition $a\delta = a\pi\mathrm{d}(s) = \mathrm{d}(s)$ is homotopic to zero. By (6) we see that the composition $\delta b = -\mathrm{d}(\pi)sb = \mathrm{d}(-\pi)$ is homotopic to zero. □

Lemma 0.4. *In Situation 0.1 let $x_i \to y_i \to z_i$ be morphisms in $\mathcal{A}$ ($i = 1, 2, 3$) such that $x_2 \to y_2 \to z_2$ is an admissible short exact sequence. Let $b : y_1 \to y_2$ and $b' : y_2 \to y_3$ be morphisms in Comp($\mathcal{A}$) such that*

$$\begin{array}{ccccc} x_1 & \longrightarrow & y_1 & \longrightarrow & z_1 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b}\downarrow & & {\scriptstyle 0}\downarrow \\ x_2 & \longrightarrow & y_2 & \longrightarrow & z_2 \end{array} \quad \textit{and} \quad \begin{array}{ccccc} x_2 & \longrightarrow & y_2 & \longrightarrow & z_2 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b'}\downarrow & & {\scriptstyle 0}\downarrow \\ x_3 & \longrightarrow & y_3 & \longrightarrow & z_3 \end{array}$$

commute up to homotopy. Then $b' \circ b$ is homotopic to 0.

Proof. By Lemma 0.2, we can replace b and b' by homotopic maps $\tilde{b}$ and $\tilde{b}'$, such that the right square of the left diagram commutes and the left square of the right diagram commutes. Say $b = \tilde{b} + d(h)$ and $b' = \tilde{b}' + d(h')$ for degree -1 morphisms h and h' in $\mathcal{A}$. Hence

$$b'b = \tilde{b}'\tilde{b} + d(\tilde{b}'h + h'\tilde{b} + h'd(h))$$

since $d(\tilde{b}) = d(\tilde{b}') = 0$, i.e. $b'b$ is homotopic to $\tilde{b}'\tilde{b}$. We now want to show that $\tilde{b}'\tilde{b} = 0$. Because $x_2 \xrightarrow{f} y_2 \xrightarrow{g} z_2$ is an admissible short exact sequence, there exist degree 0 morphisms $\pi : y_2 \to x_2$ and $s : z_2 \to y_2$ such that $\mathrm{id}_{y_2} = f\pi + sg$. Therefore

$$\tilde{b}'\tilde{b} = \tilde{b}'(f\pi + sg)\tilde{b} = 0$$

since $g\tilde{b} = 0$ and $\tilde{b}'f = 0$ as consequences of the two commuting squares. □